Certify, Then Switch: A Physics-Informed Operator for Grid Reconfiguration Under Out-of-Distribution Data-Center Loads

Sehaj Randhir Singh
Queens High School for the Sciences at York College
New York, NY
sehajrsinghs@gmail.com

Abstract

A learned grid controller is kept out of the control room for a reason that is easy to state: a predicted state that violates Kirchhoff's laws by even a tenth of a percent is not a state, because the flows it implies are carried by no equipment. We build a two-stage controller for topology reconfiguration when data-center load grows past its training envelope, and measure it where the exclusion applies. An admittance-weighted message-passing operator is trained under a Kirchhoff residual penalty evaluated on its own prediction. The operator ranks candidate line-switching actions, and a Newton-Raphson polish either certifies the chosen state or rejects the decision. At three times the largest surge in the training sample, the penalized operator leaves a residual at the ninety-fifth percentile of 34.5% of the system base where a supervised operator of identical size leaves 149.9% and a per-bus MLP leaves 114.7%. None of the three is inside the 0.1% tolerance, which is why the decision is certified rather than predicted, and the certified switch restores a feasible state on 56.9% of the tasks that admit one, against 14.8% for the supervised operator and 0% for the classical rule. Every number regenerates from committed run records.

1 Introduction

Grid load is being reshaped by two forces that pull in opposite directions.Data centers add large blocks of nearly constant demand on the timescale of a single interconnection queue, consumption projected to grow by roughly 15% per year to 2030 [IEA, 2025], while inverter-based generation adds capacity that is fast to build and volatile in output. The consequence for a learned controller is that the map from inputs to a valid operating point changes: the historical sample describes a grid whose transfer limits were set for a different load pattern, and the switch settings that relieved a corridor last year may be the ones that overload it now. That is the setting we study, as a compliance problem rather than a forecasting one.

The reason is specific. A state for a transmission network is a set of bus voltage phasors satisfying the injection equations induced by the admittance matrix, and a neural surrogate returns a vector that looks like one and generally is not. A residual of a tenth of a percent of the system base means the implied branch flows are carried by no conductor, which is small in the units a regression reports and catastrophic in the units a protection engineer uses, so accuracy is the wrong metric and the right one is the residual evaluated on the prediction itself.

The literature supplies both halves separately. Learned operators map fields to solutions of partial differential equations at high accuracy [Li et al., 2021, Kovachki et al., 2023], training on a residual is established [Raissi et al., 2019], graph surrogates predict power-flow solutions [Habib et al., 2023, Zamzam and Baker, 2020], the community scores compliance through shared benchmarks [Leyli-Abadi et al., 2022, Zheng et al., 2022], and topology control is a mixed-integer program over line

Preprint. Unpublished manuscript.

statuses [Fisher et al., 2008, Hedman et al., 2009], exact and too slow for the seconds a corrective switch has. What the learned literature has not measured is the residual outside the training envelope, and what the control literature has not built is a selector whose output physics checks first.

We build that pair and measure it. The first component is an admittance-weighted message-passing operator, whose admittance matrix supplies both the aggregation weights of the graph layers and the differential operator of the loss, evaluated on the predicted state. The second is a verifier: the operator ranks candidate actions by the state it predicts for each, and a Newton-Raphson polish from the winner either drives the residual under tolerance, making the polished state an exact power-flow solution whose feasibility is a fact, or fails, and the decision is rejected. Two results carry the paper. At three times the largest training surge the penalized operator's residual is 34.5% of the base at the ninety-fifth percentile, against 149.9% for the supervised operator and 114.7% for the per-bus MLP, and no variant is inside the tolerance, which is why the verifier exists. The verifier is then cheap and uniform: six Newton iterations certify 96.8% of the penalized operator's out-of-envelope predictions, and the operator's contribution, the starting point and the ranking, turns into a feasible state on 56.9% of recoverable tasks against a 27.4% ceiling from exhaustive search. Section 2 defines the state and the residual, Section 3 the operator and the verifier, and Section 4 the two results.

2 Problem formulation

State and residual. A network has n buses and m branches. Branch k connects bus f_k to bus t_k with series admittance y_k and total charging susceptance b_k, and the bus admittance matrix $\mathbf{Y} \in \mathbb{C}^{n\times n}$ collects them, with $\mathbf{Y}_{ik}$ the mutual admittance between buses i and k and $\mathbf{Y}_{ii}$ the sum of the admittances incident to bus i plus any bus shunt. The complex bus voltage is $V_i = V_{m,i}e^{\mathrm{j}\theta_i}$, where $V_{m,i}$ is the magnitude in per unit and θ_i the angle in radians, and a scenario specifies the active and reactive load $P_{d,i}$ and $Q_{d,i}$ in MW and MVAr, the generator dispatch P_g and voltage setpoints V_g, and the branch status $\sigma \in \{0,1\}^m$, where $\sigma_k = 0$ means branch k is switched out.

The injection a state implies at bus i is $S_i(\mathbf{V}) = V_i \sum_k \overline{\mathbf{Y}_{ik} V_k}$, and the scenario specifies $S_i^\star = (P_{g,i} - P_{d,i}) + \mathrm{j}(Q_{g,i} - Q_{d,i})$ divided by the system base S_b. A state is a power-flow solution exactly when $S_i(\mathbf{V}) = S_i^\star$ at every bus whose injection is constrained: active power at every bus except the reference, and reactive power at every bus without a voltage-controlling generator. We write the *Kirchhoff residual* of an arbitrary state as

$$\eta(\mathbf{V}) = \max\Big(\max_{i\neq\mathrm{ref}} |\Re\{S_i(\mathbf{V}) - S_i^\star\}|,\ \max_{i\in\mathcal{PQ}} |\Im\{S_i(\mathbf{V}) - S_i^\star\}|\Big), \tag{1}$$

reported as a percentage of S_b, so $\eta = 0.1\%$ is one part in a thousand of the system base. Every compliance claim in this paper is a statement about (1) on a predicted state.

A state is *feasible* when it is a solution, $\eta = 0$; when every in-service branch carries apparent power below its thermal rating, which we normalize so that a loading of 1 is the rating; and when every voltage magnitude lies in the operating band $[0.94, 1.06]$ per unit.

The surge envelope. Let B_ref be a reference data-center block, chosen per network by calibration (Appendix B), and let a scenario's block be sB_ref, split across n_{dc} sites by a Dirichlet draw so that a surge is not a uniform scaling. The training sample uses $s \in [0.15, 1]$, so $s = 1$ is the edge of the envelope and $s = 3$ is three times the largest block the operator ever saw; every out-of-distribution claim below is therefore a claim about extrapolation past the training maximum. Renewable output is drawn independently of s and injected as negative load with inverter reactive support, while the added load is served pro rata from generator headroom, so the strain lands on the transmission system rather than the dispatch stack. Contingencies and the candidate switching set are defined in the supplementary setup, with the exhaustive sweep over that set that is the ceiling every method is scored against.

3 Method

Admittance-weighted message passing. The operator maps a scenario to a voltage state. Buses carry the load and generation in per unit, bus type, the voltage limits, the site's share of the data-center block, its renewable availability, the surge multiple and the row sum of $|\mathbf{Y}|$; branches carry their per-unit impedance and charging susceptance, tap ratio, status and thermal rating. Features are embedded by two-layer perceptrons, and three message-passing layers update the node states, with the message from bus k to bus i weighted by $\alpha_{ik} = |\mathbf{Y}_{ik}| / \sum_j |\mathbf{Y}_{ij}|$ over in-service branches. The weighting is the point: the receptive field of one layer is the physical neighborhood, the weights are

network constants rather than learned attention, and the layer inherits invariance to bus relabeling that a spectral operator does not have. Two heads give the voltage magnitude and angle, bounded around nominal at 1 ± 0.2 per unit and ± 1.2 radians rather than by the case's limits, because a head bounded by the limits makes an infeasible state unrepresentable and the compliance metric vacuous. Branch flows are not predicted: they follow from the predicted state through the branch admittances, so no flow is ever predicted that the operator's own voltages do not imply.

Training on the residual. The loss is $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_k \mathcal{L}_{\text{phys}}$, where $\mathcal{L}_{\text{data}}$ is the mean squared error of the voltage magnitude and angle against the solved state, and $\mathcal{L}_{\text{phys}}$ is the squared version of (1) over the constrained buses, evaluated on the *predicted* state. The multiplier is not tuned but updated, $\lambda_{k+1} = \min(\lambda_{\max}, \lambda_k + \beta\sqrt{\mathcal{L}_{\text{phys}}})$, so it grows with the achieved violation rather than with its square and cannot saturate in one epoch; we use $\beta = 0.5$ and $\lambda_{\max} = 10^3$. Checkpoints are selected on validation physical compliance, the ninety-fifth percentile of the residual on held-out in-distribution scenarios. All four variants are trained under the identical rule, so the comparison isolates the mechanism: the supervised operator is this network with $\lambda_k \equiv 0$, and the per-bus MLP sees the same features with no graph structure.

Propose, then verify. The controller never acts on a predicted state. It builds one status matrix per candidate, runs the operator once over that batch, and scores each candidate by $\max(0, \text{loading} - 1) + \max(0, \text{voltage excursion}) + \eta$ on the predicted state, so an impossible state is never a good candidate. The winner is handed to the verifier: Newton-Raphson starts from the predicted state, takes at most six iterations, and the decision counts as certified only if the mismatch is driven under tolerance, in which case the polished state is an exact power-flow solution and its feasibility is a fact; otherwise it is rejected. The cost is one batched forward pass plus six iterations on one candidate, against a full solve per candidate for the sweep, so the operator is a warm start and a ranker and the guarantee sits where it can be honored.

4 Results

Setup. The network we report is the IEEE 14-bus case, exported from pandapower [Thurner et al., 2018] and solved by our own Newton-Raphson implementation, which agrees with pandapower to 1.3×10^{-15} per unit in voltage magnitude and 1.4×10^{-12} MVA in terminal flow over 20 single-outage states. Appendix A repeats that check on the IEEE 30- and 118-bus cases. The suite trains on 600 scenarios, validates on 200, and tests on 200 per regime over 3 seeds, and every table reports the mean over seeds with the spread.

Result 1: the residual penalty is what holds the prediction physical. The first question is whether a learned state stays a state when the load leaves the envelope, and the answer depends on the training objective rather than on the architecture, which the ablation isolates, since the supervised operator is this network with the penalty removed and the pair therefore differs in objective alone. On in-distribution scenarios the penalized operator leaves a residual at the ninety-fifth percentile of 19.3% of the system base, against 113.8% for the supervised operator and 88.7% for the MLP, and none of the three is inside the 0.1% tolerance, which the supplementary regime table reports at the same value for every variant. A tenth of a percent of the base is a demanding bar for a regression, so the informative quantity is how far above it each model sits. At the surge of three times the training maximum the penalized operator's residual is 34.5%, against 149.9% for the supervised operator, a factor of 4.3 above the penalized operator, and 114.7% for the MLP, while the penalized residual grows by 1.8 from its in-distribution value, so the penalty is a bound on how far the prediction can drift rather than a guarantee that it stays inside the tolerance. The mechanism is direct: the penalized operator is trained to satisfy the equations it will be checked on, and that constraint does not stop applying when the inputs move, whereas a model trained only to reproduce states has no term distinguishing a comfortable interpolation from an impossible extrapolation. Figure 1 shows the same comparison as a function of the surge, and it is the failure that carries the reading: the penalized curve stays within a factor of two of its in-distribution value, while the supervised curve and the MLP are already far above the tolerance inside the envelope and climb from there. The compound contingency regime in the last column of Table 3 is the harsher test, since the states it contains were never jointly observed, and it degrades every model. The same measurement bounds the claim. The penalized operator does not meet the tolerance outside the envelope either, at 100.0% of scenarios, so the honest reading is that the residual penalty reduces an arbitrary error to a bounded one and does not remove it, which is why the contribution is a pair rather than a single model. Voltage accuracy does not track compliance

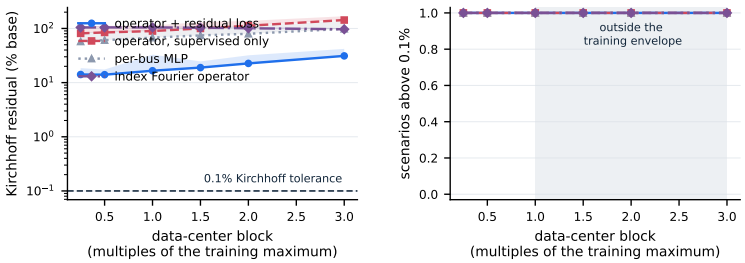


Figure 1: Physical compliance against the data-center surge multiple, for the 14-bus suite over 3 seeds and 120 scenarios per level; the shaded region is outside the training envelope. Panel (a) is the mean Kirchhoff residual of the predicted state on a logarithmic axis, the band spanning the mean to the ninety-fifth percentile, and the dashed line is the 0.1% tolerance; panel (b) is the share of scenarios above it. The four curves are the four variants of Section 3 under one checkpoint rule, so the two operators differ only in the residual penalty the comparison isolates, and the split is by scenario.

at all: at the same surge the per-bus MLP, with no graph structure and no residual term, reaches 0.010 per unit against 0.017 for the penalized operator while leaving the largest residual of the three, so a model can be the most accurate and the least physical at once.

Result 2: certification is uniformly cheap, so the operator is scored on what it is for. A prediction that is close to physical is useful only if something can finish the job, so the verifier takes the predicted state and runs a fixed Newton budget of six iterations from it, under the same solver settings and voltage band the task is scored on. On out-of-envelope scenarios the polish certifies 96.8% of the penalized operator's predictions and 96.8% of the supervised operator's, and in both cases it drives the median residual to 8.5×10^{-13}% of the base. The near-equality is a finding rather than a nuisance: a Newton polish from either starting point converges to the same solution of the same equations, so the certified state carries no trace of the model that proposed it, and the two share a post-certification feasibility rate of 32.2%. The budget is what matters on the other side of the same measurement: with a single Newton iteration the polish certifies 0% of the same predictions, so the guarantee is a statement about a budget as much as about a model, and the value the operator adds has to be looked for in the starting point, which sets whether the budget is enough, and in the ranking of candidates, which the verifier cannot check without being handed one.
The reconfiguration task is where the pair is scored as a controller, and Table 4 reports it. Between one and 2.00 times the training envelope the base state is feasible on 36.7% of tasks and the candidate set contains a feasible action on 27.4% of them, which is the target. The certified switch reaches 56.9% of that target against 14.8% for the supervised operator and 0% for the classical rule that opens the most heavily loaded circuit. The mechanism is the ranking: a candidate is chosen by the state predicted for it, so an operator whose residual is large out of distribution ranks candidates by a quantity that includes its own error, and its agreement with the exhaustive best action is 25.8% against 5.21%, and the verifier keeps that failure affordable: a rejected decision costs solver iterations rather than a blackout. At the extreme band the conclusion is bounded: the base state is feasible on 1.67% of tasks and the exhaustive sweep finds a feasible action on only 0% of them, so for most of these tasks no single switching action restores feasibility, and topology control buys margin up to roughly 2.00 times the training envelope and no more.

5 Discussion and limitations

The two results compose into one claim. A learned operator can be trained to stay close to physics outside the sample it was drawn from, that property does not extend to exactness, and exactness is what a control room needs, so the operator is placed where its error is affordable and the solver where its cost is affordable. What supports the claim is the residual, which no accuracy metric reports, and what bounds it is the exhaustive sweep, which turns a success rate into a share of what was achievable.

Limitations. The networks are academic cases, every scenario is synthetic, and ratings are synthesized where a case ships a placeholder, at so the absolute shares would move under real data. Topology control here is a single line switch, and the verifier treats a non-convergent solve as no feasible action, which is honest for a planning study and insufficient for a control room where it is itself an alarm. Frequency, protection and cascade dynamics are outside the paper, and the out-of-distribution axis is a load envelope and a compound contingency rather than a shift to an unseen network.

References

E. B. Fisher, R. P. O'Neill, and M. C. Ferris. Optimal transmission switching. *IEEE Transactions on Power Systems*, 23(3):1346–1355, 2008.

B. Habib, E. Isufi, W. van Breda, and P. Palensky. Deep statistical solver for distribution system state estimation. *arXiv preprint arXiv:2301.01835*, 2023.

K. W. Hedman, R. P. O'Neill, E. B. Fisher, and S. S. Oren. Optimal transmission switching with contingency analysis. *IEEE Transactions on Power Systems*, 24(3):1577–1586, 2009.

International Energy Agency. Energy demand from AI. In *Energy and AI*. IEA, Paris, 2025.

N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

M. Leyli-Abadi, A. Marot, and others. Learning industrial physical simulation benchmark suite. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2022.

Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

L. Thurner, A. Scheidler, F. Schäfer, J. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun. pandapower: an open-source python tool for convenient modeling, analysis, and optimization of electric power systems. *IEEE Transactions on Power Systems*, 33(6):6510–6521, 2018.

A. S. Zamzam and K. Baker. Learning optimal solutions for extremely fast AC optimal power flow. In *IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids*, 2020.

X. Zheng, N. Xu, L. Trinh, D. Wu, T. Huang, S. Sivaranjani, Y. Liu, and L. Xie. A multi-scale time-series dataset with benchmark for machine learning in decarbonized energy grids. *Scientific Data*, 9:359, 2022.

A Verification of the solver

Table 1: Independent verification of the solver that supplies every label and every certificate. Each network is compared against pandapower on an N-1 sweep over the branches whose endpoints are not shared with another circuit, with reactive-limit switching disabled on both sides so that the two solve the same equations. *States* is the number of swept states both solvers converged on, and *excl.* is the number excluded because one of them did not, so a row that failed is never counted as agreement. The next three columns are the largest disagreement over the compared states in voltage magnitude, in voltage angle and in branch apparent power, and the last column is the largest relative difference in apparent power over the branch rows carrying at least one MVA, which is the statistic a small denominator would inflate. *Base* is the power-balance residual of our solver alone on the unswept case, a convergence check rather than a comparison, reported to show that the sweep starts from a genuine solution rather than a converged-looking iterate. Parallel circuits are dropped from the sweep, since a row-wise outage of one twin is indistinguishable from the other by terminal buses and would silently redistribute flow between them.

Network	states	excl.	base (p.u.)	$\max \lvert\Delta V_m\rvert$ (p.u.)	$\max \lvert\Delta\theta\rvert$ (deg)	rel. flow (%)
IEEE 14	20	0	1.1×10^{-14}	1.3×10^{-15}	2.3×10^{-13}	1.89×10^{-11}
IEEE 30	19	1	1.4×10^{-14}	2.1×10^{-9}	1.3×10^{-7}	4.81×10^{-6}
IEEE 118	18	1	5.7×10^{-14}	2.6×10^{-7}	8.1×10^{-4}	1.15×10^{-2}

B Envelope calibration and setup

Table 2: The envelope and how serviceable it is, measured on the generator rather than on any model. The data-center block is the largest reference block whose in-distribution feasible share stays above 0.85, which keeps the training sample serviceable while making the three-times envelope genuinely stressed rather than nominally harder. Loading is the median worst in-service branch loading at that surge, where 1.0 is the thermal rating, and the last column is the same quantity at three times the envelope.

Network	block (MW)	feasible at 1×	loading at 1×	feasible at 3×	loading at 3×
IEEE 118	848	0.00	3.09	0.00	9.07
IEEE 14	26	0.85	0.84	0.00	1.36

Contingencies, and the task. Training contains single outages from a credible list, the branches whose outage leaves a feasible network at the nominal point. Testing adds compound contingencies drawn from the dozen most heavily loaded branches, which never occur together in training. A task adds a candidate set $\mathcal{A}$ of one switch or none, and a method succeeds when the action it selects leads, after verification, to a feasible state. Because $\mathcal{A}$ is finite it is swept exhaustively, which is the ceiling every method is scored against.

C Full regime table and per-seed records

D Scaling the envelope to a larger network

Why the reported suite is the 14-bus one is a property of the generator rather than of the network size, and it is worth recording because it is the first thing that went wrong. The IEEE 118-bus case with the default envelope declares a data-center block of 20% of base load, and the training regime that results is unusable: at the training maximum surge, the share of feasible scenarios is 0% and the median worst-branch loading is 3.09 of rating, before any surge beyond the envelope is considered. The renewable fleet is what does it. An inverter fleet sized at the default share and concentrated at three buses overloads a corridor by itself, so the block size is not the binding parameter and sweeping it alone reports the wrong culprit; reducing the block and re-probing leaves the regime unserviceable for the same reason. Calibrating a network therefore means sweeping both shares together, which is what `tools/calibrate.py` does, and the 118-bus scaling study starts from that calibration rather than from a preset.

Broader impact

The pathway this work supports is concrete. Interconnection requests and transfer limits are approved against a feasibility study, and a certified action is what lets an operator act inside a margin a study has already granted, for instance by energizing a data-center block whose transient demand exceeds

Table 3: Physical compliance by regime, mean over seeds with the spread across seeds. The residual is the maximum Kirchhoff mismatch of the predicted state, as a percentage of the system base, so 0.1% is one part in a thousand, and the two entries per cell are the mean and the spread across seeds. The share of scenarios above that tolerance is 100 percent for every variant in every regime, so the tolerance is missed everywhere and the table reports the residual, which is the quantity that still separates the variants. All learned variants share the same features, depth and channel width, and every checkpoint is chosen by validation compliance, so the rows differ in training objective only, and the fourth row changes the architecture while holding the training rule fixed.

Model	in distribution	3x load, same grid	compound outage	3x load and compound outage
operator + residual loss (ours)	19.29±2.21	34.48±4.03	111.89±13.94	117.65±20.35
operator, supervised only	113.83±43.70	149.92±39.77	197.85±10.88	211.20±14.19
per-bus MLP	88.65±1.16	114.70±9.89	218.31±6.60	251.98±9.41
index Fourier operator	111.35±0.88	107.69±0.86	110.02±0.61	107.27±1.46

Table 4: Reconfiguration performance, mean over seeds with the spread across seeds. A task is recoverable when the exhaustive sweep over the same candidate set finds a feasible action, and the ceiling row is that sweep, so the other rows are read as a share of what was achievable rather than against an assumed bound. Certified is the share of decisions that survived the Newton polish, and the time is the wall clock per decision, which includes the operator's forward pass over all candidates and the verification of the chosen one.

	surge 1.0–1.8x			surge 2.4–3.0x		
Method	recoverable	certified	ms	recoverable	certified	ms
no action	0.00±0.00	0.00±0.00	0±0	—	0.00±0.00	0±0
open the loaded line	0.00±0.00	0.89±0.04	0±0	—	0.98±0.02	0±0
per-bus MLP	0.06±0.09	0.89±0.04	0±0	—	0.98±0.02	0±0
operator, supervised only	0.15±0.21	0.90±0.04	0±0	—	0.98±0.02	0±0
operator + residual loss (ours)	0.57±0.09	0.93±0.04	0±0	—	0.97±0.03	0±0
exhaustive sweep (ceiling)	0.27±0.02	—	0±0	0.00±0.00	—	0±0
base state feasible (no action)	0.43±0.05			0.02±0.01		

the sample any historical model was fitted to. The stage is equally concrete and more modest: a certificate from a Newton polish on a synthetic 14-bus model, for one action at a time, is a statement about a solved equation and not an authorization to operate a transmission system. Deployment would require the contingency list to be the operator's own, the ratings to be conductor ratings, and the certificate to cover the actions the operator is permitted to take.

Use of AI tools

Generative AI tools were used to draft and edit this manuscript and to help implement the experimental code. Every number is substituted into the text from the committed per-seed run records by `tools/fill_numbers.py`, which fails rather than leaving a value unset, and no quantitative claim was written by hand from any other source. All cited references were checked against their published records.

Table 5: Per-seed residuals, ninety-fifth percentile as a percentage of the system base, for the operator trained with and without the residual penalty, which is the pair the claims compare. Reported so that the spread behind the mean is auditable rather than asserted.

Regime	Model	seed 0	seed 1	seed 2
in distribution	operator + residual loss (ours)	18.980	22.140	16.757
in distribution	operator, supervised only	175.583	85.174	80.747
3x load, known topology	operator + residual loss (ours)	36.678	37.943	28.828
3x load, known topology	operator, supervised only	183.938	94.125	171.696
in-envelope load, compound outage	operator + residual loss (ours)	113.733	93.962	127.960
in-envelope load, compound outage	operator, supervised only	190.784	189.539	213.216
3x load and compound outage	operator + residual loss (ours)	122.939	90.499	139.501
3x load and compound outage	operator, supervised only	200.841	201.482	231.267

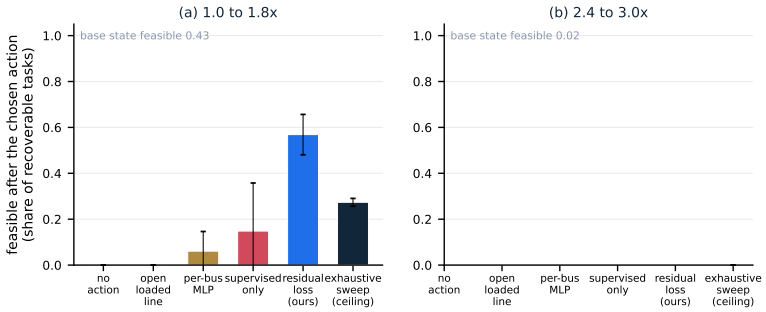


Figure 2: Verified feasibility recovery on the reconfiguration task, for the 14-bus suite over 3 seeds, with 38 tasks in the moderate band and a larger set in the extreme band. Each bar is the share of tasks that admit a feasible action, as established by the exhaustive sweep, on which the method's chosen action is feasible after certification; the ceiling is that sweep itself. Panel (a) is the moderate band, data-center surge between one and 2.00 times the training maximum, where the base state is feasible on 36.7% of tasks. Panel (b) is the extreme band, surge between 2.40 and three times the training maximum, where the base state is feasible on 1.67% of tasks and the candidate set itself contains a feasible action on 0% of them, so the shortfall against the ceiling is a limit of topology control rather than of the methods.

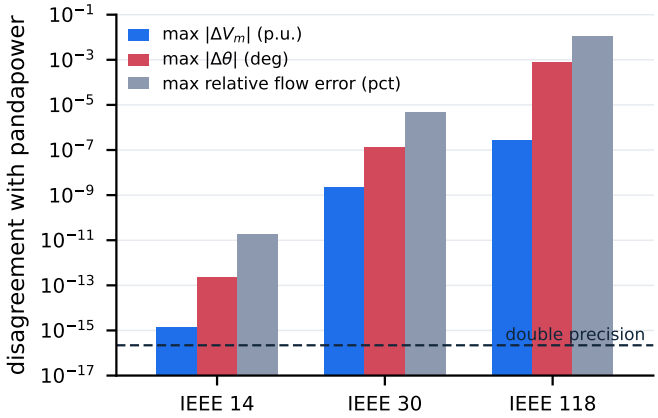


Figure 3: Disagreement between our Newton-Raphson solver and pandapower on the N-1 sweep over the largest parallel-free subset of branches, with the configuration matched by disabling reactive-limit switching on both sides. The comparison covers 20, 19 and 18 swept states on the 14-, 30- and 118-bus cases. The disagreement grows with network size, and the largest relative branch-flow difference stays below one percent, at 0 of the compared rows above that level. The three series carry different units, so the axis compares the scales at which the two solvers agree rather than like quantities, and the dotted line is the round-off floor of double precision, which the 14-bus voltage error lies at. This is the artifact behind the claim that the verifier is an independent solver rather than a second learned model.

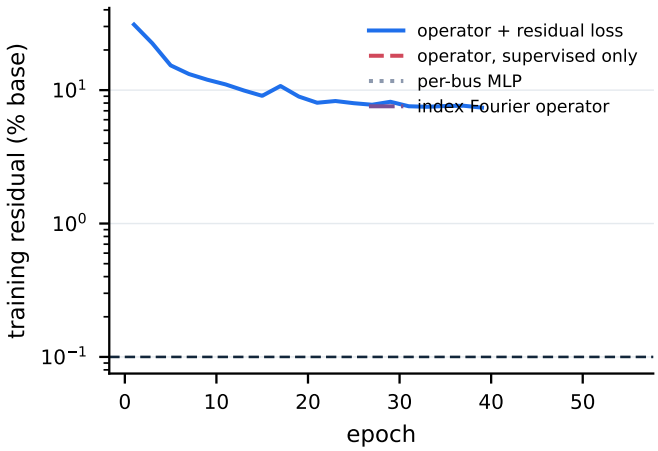


Figure 4: Training residual against epoch for one seed of the reported suite, for each learned variant, reported as the root mean square injection mismatch in percent of the system base. The dashed line is the 0.1% tolerance, which no curve reaches: this is the training diagnostic, and the compliance the paper reports is measured on held-out scenarios and on the two out-of-envelope regimes. The supervised and per-bus variants are shown under the same optimizer, schedule and checkpoint-selection rule as the penalized operator, so a curve that sits lower here is not the mechanism the paper isolates.